CAMPUS NAVIGATION AND EVENT MANAGEMENT SYSTEM DEMONSTRATIONS

ENSF 338

Term Project

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DEMO SCENARIOS AND SCREENSHOTS

Shortest Path Query

For the following queries, the campus data is imported from a file that contains different campus buildings, and ideally accurate distances between some of them.

The queries use the shortest path algorithm for the campus graph, which implements Dijkstra's algorithm and returns the total walking time and a stack with the route location ordering.

Biological Sciences to Kinesiology Block A

```
Enter the id of the starting point: BI
Enter the id of the end point: KNA

The shortest route, with a total walking time of 7.59 minutes is:
BI (Biological Sciences) -> SA (Science A) -> MH (MacEwan Hall) -> MSC (MacEwan Student Centre) -> KNA (Kinesiology Block A)
```

Schulich School of Engineering to Administration

```
Enter the id of the starting point: EN
Enter the id of the end point: AD

The shortest route, with a total walking time of 6.04 minutes is:
EN (Schulich School of Engineering) -> MH (MacEwan Hall) -> HNSC (Hunter Student Commons) -> AD (Administration)
```

Undo Navigation

```
Enter the id of the starting point: MS
Enter the id of the end point: AD

The shortest route, with a total walking time of 2.76 minutes is:
MS (Math Sciences) -> SA (Science A) -> AD (Administration)
```

```
------------------------------------------------------------

Choose one of the following operations:
        1. View buildings
        2. View route history
        3. View bookings
        4. Make service request
        5. Exit
Enter your choice by number: 2

------------------------------------------------------------

Current System State: Last navigated to AD from MS.
Stored History States: 1

------------------------------------------------------------

Choose one of the following operations:
        1. Undo last route
        2. Exit
Enter your choice by number: 1

Action: Reverted navigation query from MS to AD.
System State Restored. History is now empty.

Press enter to continue.

------------------------------------------------------------

Navigation history is empty. No previous routes available.

Press enter to return to the main menu.

------------------------------------------------------------

Choose one of the following operations:
        1. View buildings
        2. View route history
        3. View bookings
        4. Make service request
        5. Exit
Enter your choice by number: 5
```

Booking range query

```
Enter Building ID to check: ict
Enter Room ID: ict-121

Enter date to view (YYYY-MM-DD):
> 2026-04-25

---------------------------------------------

Bookings for ICT-121 on 2026-04-25:
- ENSF 338 Workshop: 10:30 to 12:00
```

Show all event for a specific room on a specific day

Priority Queue Demo

Enqueue Operations

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 1

Enter request: request1

------------------------------------------

Select request priority:
        1. Emergency
        2. Standard
        3. Low
Enter priority by number: 2
Request queued with Standard priority.
```

1: First Standard Priority Enqueue

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 1

Enter request: request2

------------------------------------------

Select request priority:
        1. Emergency
        2. Standard
        3. Low
Enter priority by number: 1
Request queued with Emergency priority.
```

2: First Emergency Priority Enqueue

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 1

Enter request: request3

----------------------------------------

Select request priority:
        1. Emergency
        2. Standard
        3. Low
Enter priority by number: 3
Request queued with Low priority.
```

3: First Low Priority Enqueue

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 1

Enter request: request4

----------------------------------------

Select request priority:
        1. Emergency
        2. Standard
        3. Low
Enter priority by number: 2
Request queued with Standard priority.
```

4: Second Standard Priority Enqueue

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 1

Enter request: request5

----------------------------------------

Select request priority:
        1. Emergency
        2. Standard
        3. Low
Enter priority by number: 1
Request queued with Emergency priority.
```

5: Second Emergency Priority Enqueue

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 1

Enter request: request6

----------------------------------------

Select request priority:
        1. Emergency
        2. Standard
        3. Low
Enter priority by number: 3
Request queued with Low priority.
```

6: Second Low Priority Enqueue

View Request Queue

```
Current Request Queue (from most to least urgent):
1. [Emergency] request2
2. [Emergency] request5
3. [Standard] request1
4. [Standard] request4
5. [Low] request3
6. [Low] request6
```

7: Requests are displayed in the correct order of priority, followed by insertion order

Dequeue Operations

Below, we can see all six dequeue operations happen in the correct order, as displayed in the previous list.

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

Now serving the highest-priority request:
Priority: Emergency
Request: request2
```

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

Now serving the highest-priority request:
Priority: Emergency
Request: request5
```

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

Now serving the highest-priority request:
Priority: Standard
Request: request1
```

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

Now serving the highest-priority request:
Priority: Standard
Request: request4
```

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

Now serving the highest-priority request:
Priority: Low
Request: request3
```

```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

Now serving the highest-priority request:
Priority: Low
Request: request6
```

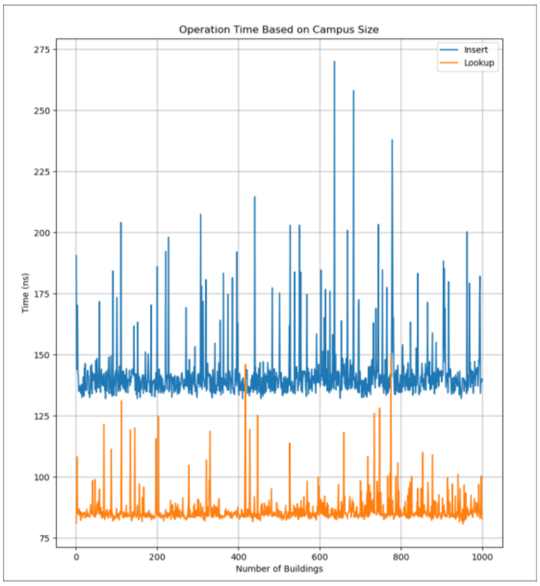
```
Choose one of the following operations:
        1. Create new request
        2. Get next Request
        3. Exit
Enter your choice by number: 2

There are no requests in the queue
```

8: Cannot dequeue if the queue is empty

Fast Lookup Demo

Insert and Lookup Time Complexity Demonstration



It is clear here that insertion and lookup time does not depend on campus size, demonstrating the O(1) behaviour.

Lookup of Key That Does Not Exist

```
Choose one of the following operations:
        1. Find shortest route between buildings
        2. Lookup building by id
        3. Add building
        4. Exit
Enter your choice by number: 2

Enter the id of the building: Test

No building found with id TEST
```

This demonstrates that a lookup of an invalid key is correctly handled.

Request Pipeline Demo

```
-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 1

Ticket "1" queued.

-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 2

Ticket "2" queued.

-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 3

Ticket "3" queued.

-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 4

Ticket "4" queued.

-------------------------------------------------------------------------------
```

Enqueuing tickets in order (1,2,3,4)

```
--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 5

Ticket "5" queued.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 6

Ticket "6" queued.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 7

Ticket "7" queued.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 8

Ticket "8" queued.

--------------------------------------------------------------------------------
```

Enqueueing tickets in order (5,6,7,8)

```
--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 9

Ticket "9" queued.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 1

Enter ticket: 10

Ticket "10" queued.

--------------------------------------------------------------------------------
```

Enqueuing tickets in order (9,10)

```
--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "1"

Press enter to continue.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "2"

Press enter to continue.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "3"

Press enter to continue.

--------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "4"

Press enter to continue.

--------------------------------------------------------------------------------
```

Dequeuing tickets (1,2,3,4)

```
-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "5"

Press enter to continue.

-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "6"

Press enter to continue.

-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "7"

Press enter to continue.

-------------------------------------------------------------------------------

Choose one of the following operations:
        1. Queue new ticket
        2. Dequeue ticket
        3. Exit
Enter your choice by number: 2

Ticket: "8"

Press enter to continue.

-------------------------------------------------------------------------------
```

Dequeuing tickets (5,6,7,8)

```
--------------------------------------------------------------------------------

Choose one of the following operations:
	1. Queue new ticket
	2. Dequeue ticket
	3. Exit
Enter your choice by number: 2

Ticket: "9"

Press enter to continue.

--------------------------------------------------------------------------------

Choose one of the following operations:
	1. Queue new ticket
	2. Dequeue ticket
	3. Exit
Enter your choice by number: 2

Ticket: "10"

Press enter to continue.

--------------------------------------------------------------------------------
```

Dequeuing tickets (9, 10)